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Degree on Mathematics
Final Degree Project

Foundations of Morse Theory: From Sard's Theorem to Morse Inequalities

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Introduction

Morse theory

$M \equiv$ Smooth manifold (compact)

$f : M \rightarrow \mathbb{R} \equiv$ Morse function

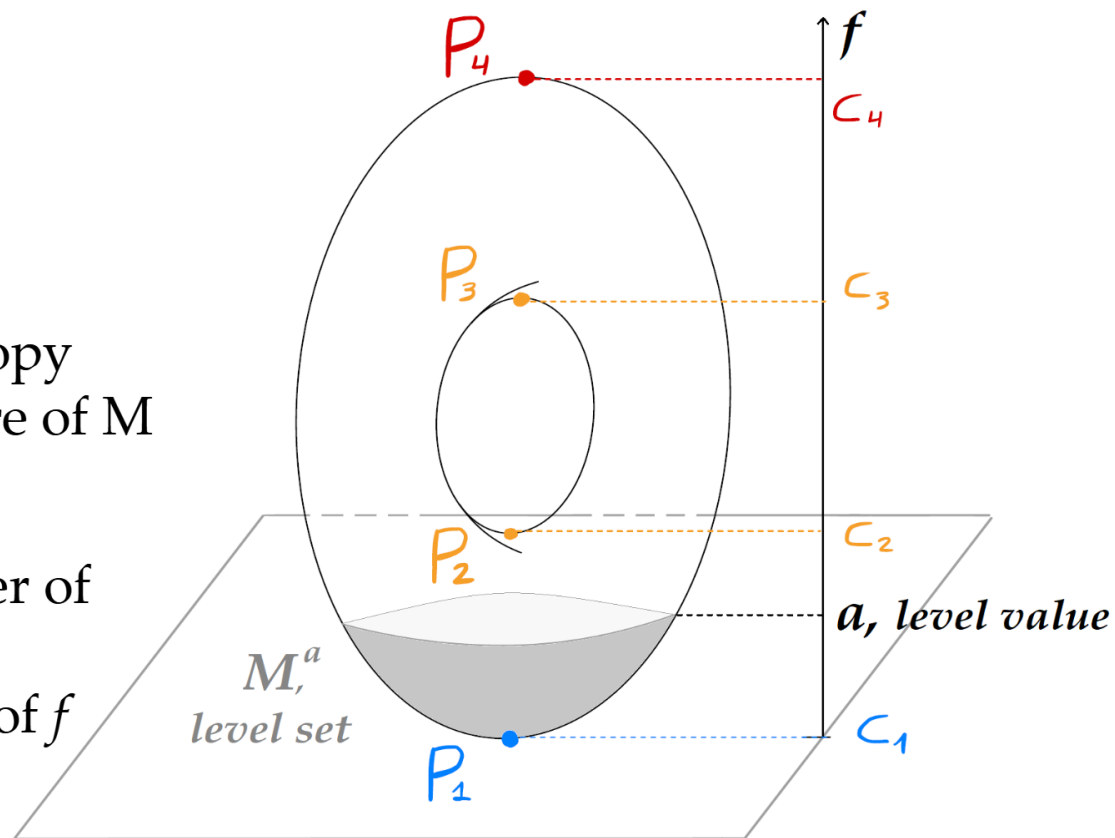
$M^a = \{p \in M : f(p) \leq a\} \equiv$ Level set

Critical points of $f \xRightarrow{\text{Morse's theorems}} \text{Homotopy structure of } M$

Different perspective:
Homotopy invariants of $M \xRightarrow{\text{Morse inequalities}} \text{Number of critical points of } f$

Example:

$M = \text{Torus}$ $f = \text{Height function}$



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1. What is a Morse Function?

$M \equiv$ Manifold, dimension m

$C^\infty(M) \equiv$ $\mathbb{R}$ -algebra of smooth functions $f : M \rightarrow \mathbb{R}$

Defintions

- If $d_p f \equiv 0$, then p is a **critical point** with associated **critical value** $c = f(p)$.
- The **Hessian** of f at the critical point p is the functional $H_p : T_p M \times T_p M \rightarrow \mathbb{R}$ with local expression $\left(\frac{\partial^2 f}{\partial x_i \partial x_j}(p) \right)$.

1. What is a Morse Function?

- The **index** of f at p is the maximal dimension of a subspace of T_pM on which H_p is negative definite.
- A critical point p is **non-degenerate** if $\left(\frac{\partial^2 f}{\partial x_i \partial x_j}(p)\right)$ is invertible.
- A **Morse Function** is a smooth function $f \in C^\infty(M)$ whose critical points are all non-degenerate.

2. Whitney's and Sard's Theorems

Whitney's Theorem: Let M be a compact manifold. There exists $n \in \mathbb{N}$ such that M is a differential submanifold of the Euclidean space $\mathbb{R}^n$.

Sard's Theorem: Let M and N be manifolds and $F : M \rightarrow N$ be a smooth function. The set of critical values of F has measure zero in N .

Idea of the proof:

1) $\left. \begin{array}{l} \psi : U \subseteq M \rightarrow \mathbb{R}^m \\ \phi : V \subseteq N \rightarrow \mathbb{R}^n \end{array} \right\}$ We need to prove that the set of critical values of $f = \phi \circ F \circ \psi^{-1}$ has measure zero in $\mathbb{R}^n$.

2. Whitney's and Sard's Theorems

2) Let $f : U \subseteq \mathbb{R}^m \rightarrow \mathbb{R}^n$ and let C be the set of its critical points.

Let $C_i = \{x \in U : f^{(k)}(x) = 0 \text{ for all } 1 \leq k \leq i\}$.

For any $k \in \mathbb{N}$:

$$C = (C - C_1) \cup (C_1 - C_2) \cup \cdots \cup (C_{k-1} - C_k) \cup C_k$$

3) It is sufficient to prove:

- | | | |
|---|---|-------------------------------------|
| i) $f(C - C_1)$ has measure zero | } | Induction on m + Fubini's Theorem |
| ii) $f(C_i - C_{i+1})$ has measure zero | | |
| iii) There exists a k such that $f(C_k)$ has measure zero | } | Taylor's Theorem |

3. Properties of Morse Functions

Abundance of Morse Functions: Let M be a compact smooth manifold embedded in the Euclidean space $(\mathbb{R}^n, \|\cdot\|)$.

For almost every point $p \in \mathbb{R}^n$, the function $f_p \in C^\infty(M)$ defined below is a Morse function:

$$f_p(q) = \|q - p\|^2.$$

Idea of the proof:

1) Define:

$$P := \{p \in \mathbb{R}^n : f_p \text{ is not a Morse function}\}$$

which is precisely the set of points p such that exists a q degenerate critical point.

3. Properties of Morse Functions

2) Apply Sard's theorem to prove that P has measure zero:

i) Consider the orthogonal fiber bundle:

$$N := \{(q, v) \in M \times \mathbb{R}^n : v \perp T_q M\}$$

ii) Define a smooth function $E : N \rightarrow \mathbb{R}^n$ as:

$$E(q, v) := q + v$$

iii) Note that $p = E(q, v)$ is the critical point condition for q . In fact, the set of critical values of E is P , since the non-invertibility of the differential of E is equivalent to

$$\frac{\partial^2 f_p}{\partial x_j \partial x_i}(q) = 2\left(\frac{\partial q}{\partial x_j} \cdot \frac{\partial q}{\partial x_i} + (q - p) \cdot \frac{\partial^2 q}{\partial x_j \partial x_i}\right) \quad \text{being non-invertible} \quad (q \text{ degenerate})$$

3. Properties of Morse Functions

Density of Morse functions: Let M be a compact smooth manifold. Then, Morse functions are dense in $C^\infty(M)$ with respect to the C^∞ topology. That is, any smooth function in $C^\infty(M)$ can be approximated arbitrarily closely in the C^∞ topology by Morse functions.

Idea of the proof:

1) Embed M into Euclidean space using a function:

$$h(q) := (f(q), h_2(q), \dots, h_n(q))$$

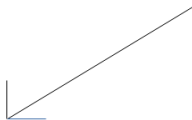


$$f \in C^\infty(M)$$

3. Properties of Morse Functions

2) Consider the Morse functions:

Perturbation of $(-c, 0, \dots, 0)$
for $c > 0$.



$$f_p(q) = \|h(q) - (-c + \varepsilon_1, \varepsilon_2, \dots, \varepsilon_n)\|^2$$
$$g(q) := \frac{f_p(q) - c^2}{2c}$$

3) Expand:

$$g(q) = f(q) + \frac{1}{c}l(q, \varepsilon_i) + \sum \varepsilon_i^2 - \varepsilon_1$$

where $l \in C^\infty(M)$, and take sufficiently large c and small ε_i .

4. Morse's Lemma

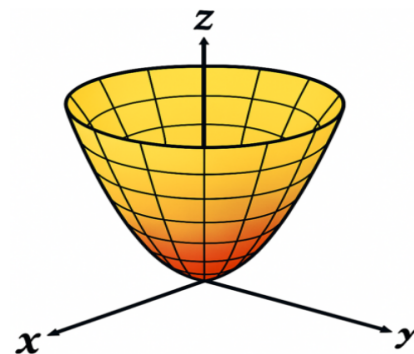
Morse's Lemma: Let M be a manifold, $f \in C^\infty(M)$ and $p \in M$ be a non-degenerate critical point of f .

There exists a local coordinate system $(y_1, \dots, y_m)$ centred at p such that the local expression of f in that system is

$$f(y_1, \dots, y_m) = f(0) - y_1^2 - \dots - y_\lambda^2 + y_{\lambda+1}^2 + \dots + y_m^2$$

where λ is the index of f at p .

Corollary: Non-degenerate critical points are isolated, and there are finitely many in a compact manifold.



5. Morse's Theorems

Definition: We define the **level set** of $f \in C^\infty(M)$ with a **level value** $a \in \mathbb{R}$ as:

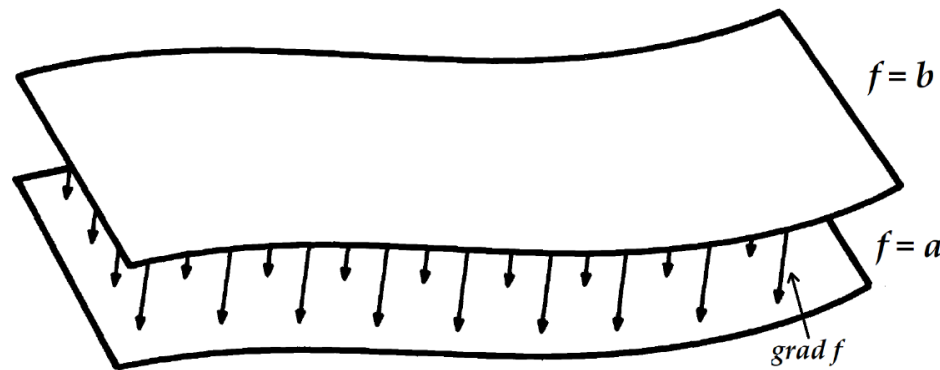
$$M^a := f^{-1}(-\infty, a] = \{p \in M : f(p) \leq a\}.$$

First Theorem of Morse: Let M be a manifold and let $f \in C^\infty(M)$. Let $a < b$ be real values, and suppose that $f^{-1}[a, b]$ is compact and has no critical points of f . Then, M^a is diffeomorphic to M^b . Furthermore, M^a is a deformation retract of M^b , so that the inclusion map $M^a \hookrightarrow M^b$ is a homotopy equivalence.

5. Morse's Theorems

Idea of the proof:

Let $(M, \langle \cdot, \cdot \rangle)$ be a Riemannian manifold



$$\langle X, \text{grad } f \rangle = X(f) \quad \forall X \in \mathcal{X}(M)$$

1) Observe:
 $C := f^{-1}[a, b]$ compact
 contained in an open set V
 (without critical points)



$\exists \gamma : M \rightarrow [0, 1]$ plateau function
 associated to (V, C)



$\text{grad } f \neq 0$ in V

5. Morse's Theorems

2) Then, $\rho := \begin{cases} \langle \text{grad } f, \text{grad } f \rangle^{-1} \cdot \gamma, & p \in V \\ 0, & p \notin V \end{cases}$ is well defined.

3) Consider the vector field $X := \rho \text{ grad } f$. Vanishes outside a compact set
 $\Rightarrow$ Generates a unique uniparametric group of diffeomorphisms:

$$\varphi : \mathbb{R} \times M \rightarrow M$$

$$(t, p) \rightarrow \varphi(t, p) = \varphi_t(p)$$

Satisfies: $f(\varphi_t(p)) = t + f(p)$

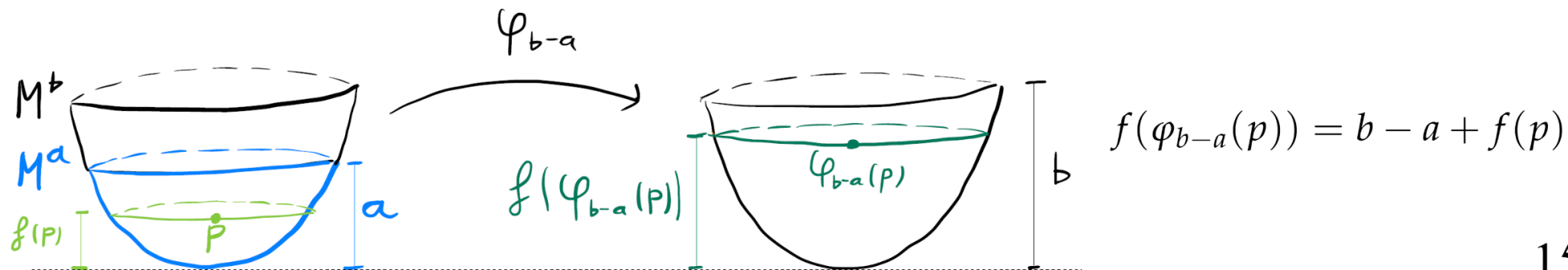
5. Morse's Theorems

4) For $t = b - a$, φ_{b-a} carries M^a diffeomorphically onto M^b .

5) We have a homotopy $r_t : M^b \rightarrow M^b$ defined as

$$r_t(p) = \begin{cases} p & \text{if } f(p) \leq a \\ \varphi_{t(a-f(p))}(p) & \text{if } a \leq f(p) \leq b \end{cases}$$

Between $r_0 = id_{M^b}$ and the deformation retraction of M^b onto M^a , r_1 .

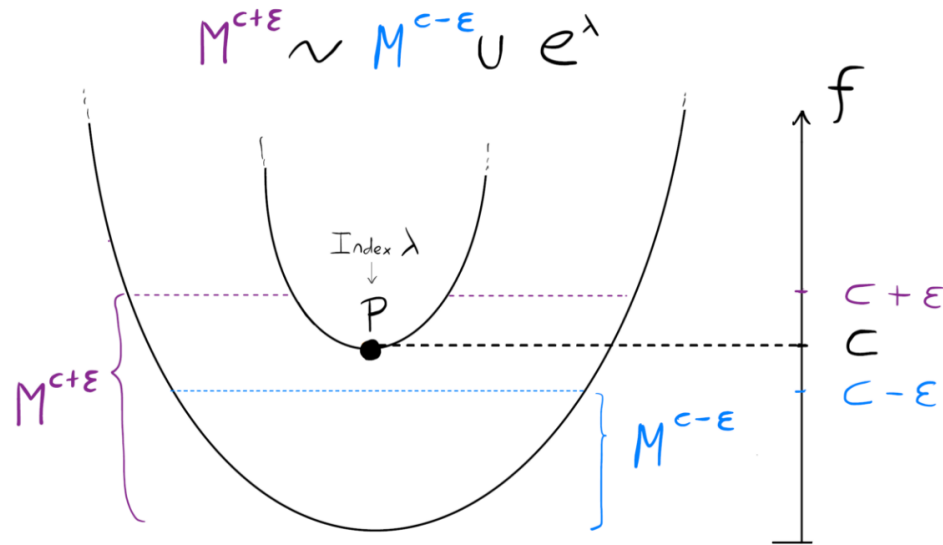


5. Morse's Theorems

Second Theorem of Morse: Let M be a manifold, $f \in C^\infty(M)$ and let $p \in M$ be a non-degenerate critical point of f with index λ .

Let $f(p) = c$ and suppose there exists a $\varepsilon > 0$ such that $f^{-1}[c - \varepsilon, c + \varepsilon]$ is compact and contains no critical points of f other than p .

Then, for all sufficiently small ε , the set $M^{c+\varepsilon}$ has the homotopy type of $M^{c-\varepsilon}$ with a λ -cell $e^\lambda \cong \{x \in \mathbb{R}^\lambda : \|x\| \leq 1\}$ attached.



5. Morse's Theorems

Idea of the proof:

1) Introduce $F \in C^\infty(M)$:

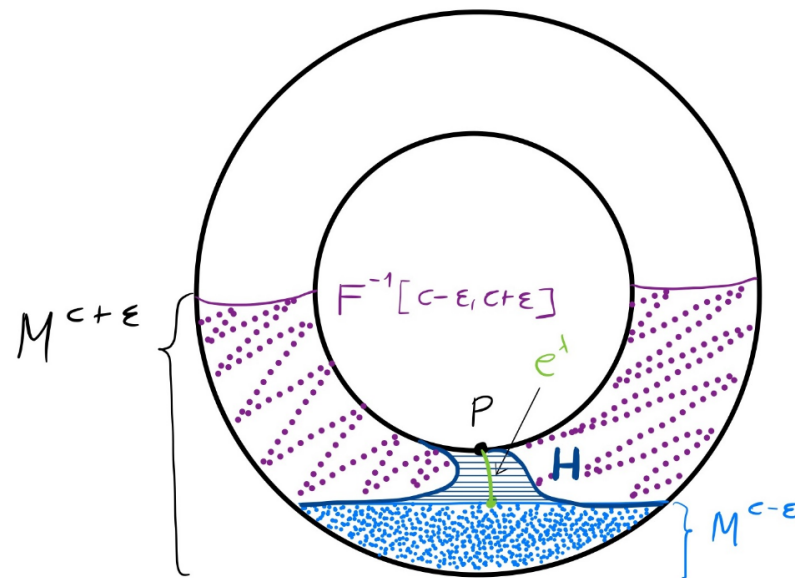
- $F < f$ in a neighbourhood of p ,
- $F = f$ otherwise,

such that $F^{-1}(-\infty, c - \varepsilon] = M^{c-\varepsilon} \cup H$.

2) Prove $F^{-1}(-\infty, c + \varepsilon] = M^{c+\varepsilon}$.

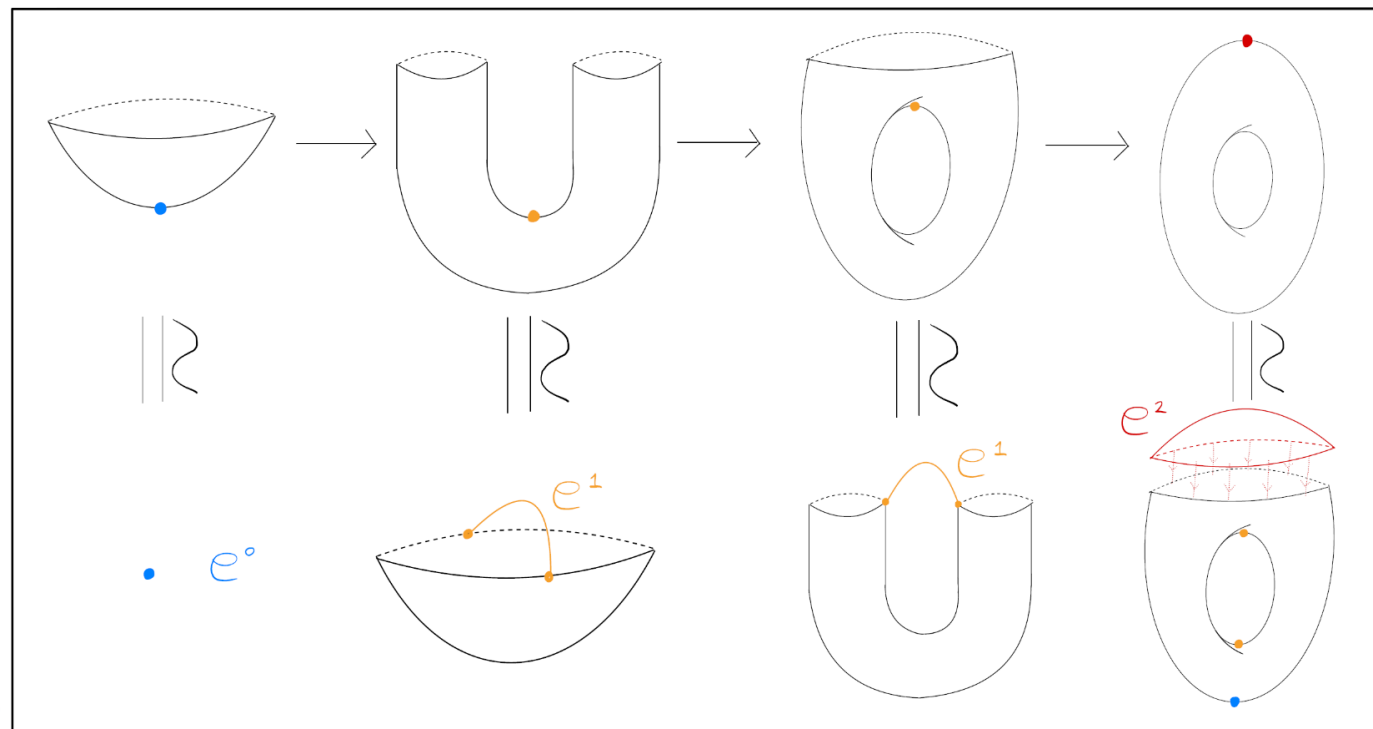
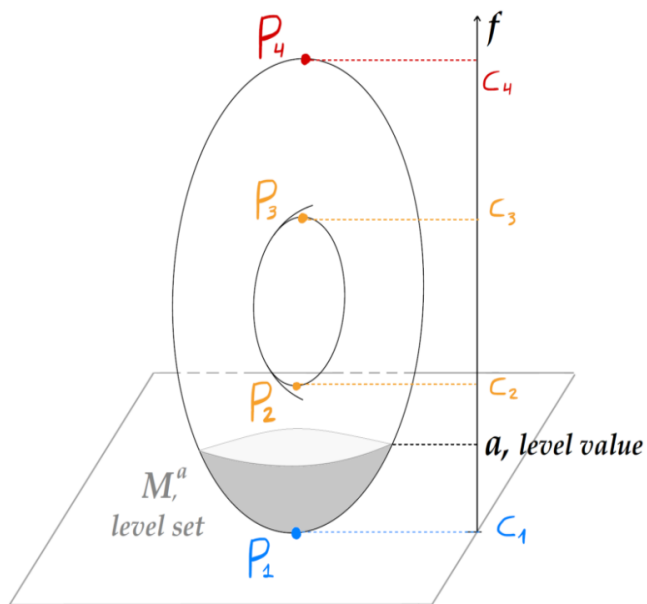
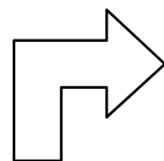
3) Apply First Theorem of Morse to $F^{-1}[c - \varepsilon, c + \varepsilon]$ to obtain that $M^{c-\varepsilon} \cup H$ is a deformation retract of $F^{-1}(-\infty, c + \varepsilon]$.

4) Consider a λ -cell $e^\lambda \subseteq H$, which contains p and is a deformation retract of H , such that $M^{c-\varepsilon} \cup e^\lambda$ is a deformation retract of $M^{c-\varepsilon} \cup H$.



5. Morse's Theorems

Example:



5. Morse's Theorems

Third Theorem of Morse: Let M be a compact manifold and $f \in C^\infty(M)$ be a Morse function. Then, M has the homotopy type of a finite CW-complex with one λ -cell attached for each critical point of index λ .

6. Morse Inequalities

Topological invariants:

- $H_n(A, B; G)$ n^{th} singular relative **homology group** of (A, B) with coefficients in G .

$$\begin{array}{ccc}
 H_n(A, B; \mathbb{Z}) \text{ finitely generated} & \xRightarrow{\text{(UCT for Homotopy)}} & H_n(A, B; F) \text{ homology} \\
 & & F\text{-vector space } (F \text{ field})
 \end{array}$$

- $b_n(A, B; F) := \dim_F(H_n(A, B; F))$ n^{th} **Betti number** of (A, B) with coefficients in F .
- $\chi(A, B; F) := \sum_{n=0}^{\infty} (-1)^n b_n(A, B; F)$ **Euler characteristic** of (A, B) with coefficients in F .

6. Morse Inequalities

Morse Inequalities: Let M be a compact manifold and $f \in C^\infty(M)$ be a Morse function. Let $c_\lambda(f)$ be the number of critical points of f with index λ , $b_\lambda(M)$ be the Betti numbers, and $\chi(M)$ be the Euler characteristic. Then:

1. **Weak Morse inequalities:**

$$b_\lambda(M) \leq c_\lambda(f), \quad \text{and} \quad \chi(M) = \sum_{\lambda=0}^{\infty} (-1)^\lambda c_\lambda(f).$$

2. **Strong Morse inequalities:**

$$b_\lambda(M) - b_{\lambda-1}(M) + \cdots \pm b_0(M) \leq c_\lambda(f) - c_{\lambda-1}(f) + \cdots \pm c_0(f).$$

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